INVERTIS UNIVERSITY, BAREILLY

FACULTY OF ENGINEERING & TECHNOLOGY

OFFICIAL ACADEMIC LEARNING RESOURCE â¢ DEPARTMENT OF CSE

Course: [BCS-301] Data Structures

Programme: Bachelor of Technology in Computer Science & Engineering **Resource Type: LECTURE NOTES**

Module / Unit: Unit 1 **Faculty Author: Dr. R. K. Sharma (Associate P**

Unit 1 Lecture Notes: Arrays, Stacks, Queues & Recursion

==========

1. Abstract Data Types & Asymptotic Complexity Analysis

- Definition of ADT: Mathematical model for data types with user-defined operations.
- Asymptotic Notations: Big-O (upper bound), Omega (lower bound), Theta (tight bound).
- Space-Time Tradeoff: Balancing memory overhead vs cpu cycles in linear data structures.

2. Stacks: Implementation, Applications & State Diagrams

- LIFO Discipline: Push, Pop, Peek, IsEmpty operations with O(1) time complexity.
- Infix to Postfix Conversion: Operator precedence parsing and Shunting-Yard algorithm.
- Function Call Stack: Activation records, local variable frames, recursion mechanics.

3. Linear & Circular Queues, Deques & Priority Queues

- Circular Queue: Overcoming false overflow via modulo arithmetic ((rear + 1) % MAX).
- Double-Ended Queues (Deque): Input-restricted and output-restricted configurations.
- Priority Queue Foundations: Heap order property preview and array-based indexing.